PLASMA PHYSICS & TOKAMAK CONFINEMENT — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: March 29, 2026

Diagram-based questions on Plasma Physics & Tokamak Confinement

Question 1

10.0 points

Analyze the toroidal and poloidal magnetic field geometry in the tokamak cross-section shown in the diagram below. Describe how these fields collaborate to confine plasma, and identify any potential areas of instability within the configuration.

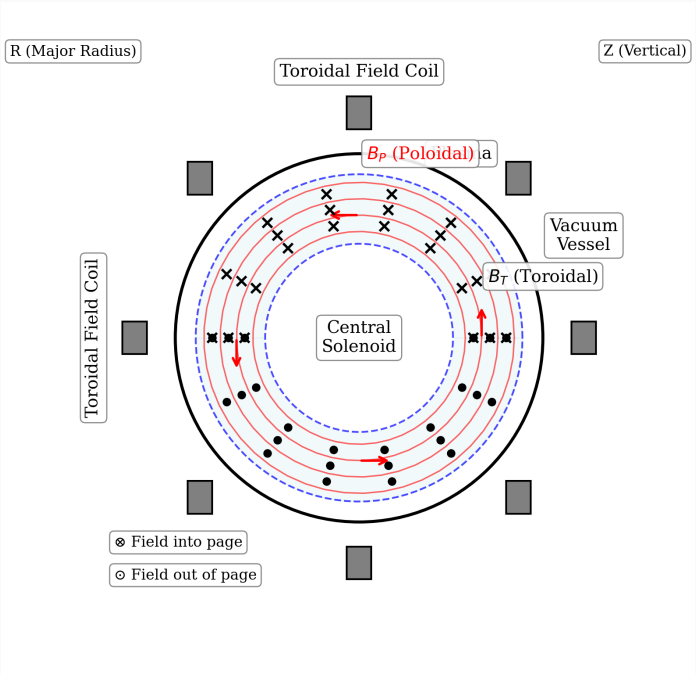

Fig. 1. Diagram for Question 1

Question 2

10.0 points

Analyze the Lawson criterion diagram below that displays the triple product $nT\tau$ versus temperature. Explain how this diagram is used to evaluate the feasibility of fusion reactions.

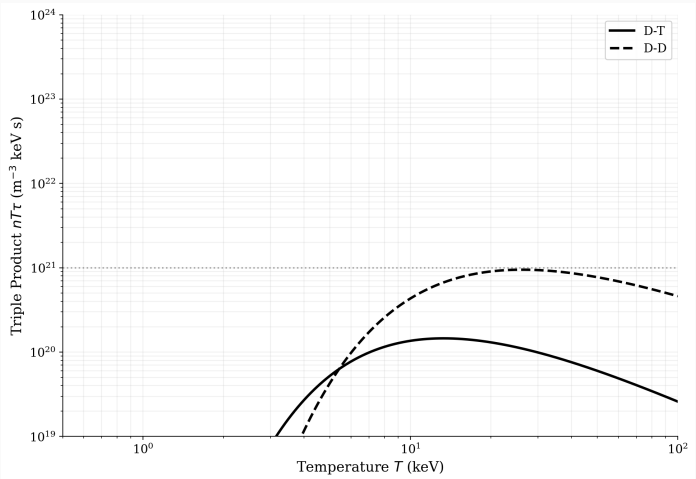

Fig. 2. Diagram for Question 2

Question 3

10.0 points

Examine the diagram below illustrating magnetic field curvature and grad-B drift for a charged particle in a tokamak. Identify how these drifts affect particle confinement and propose mitigation strategies.

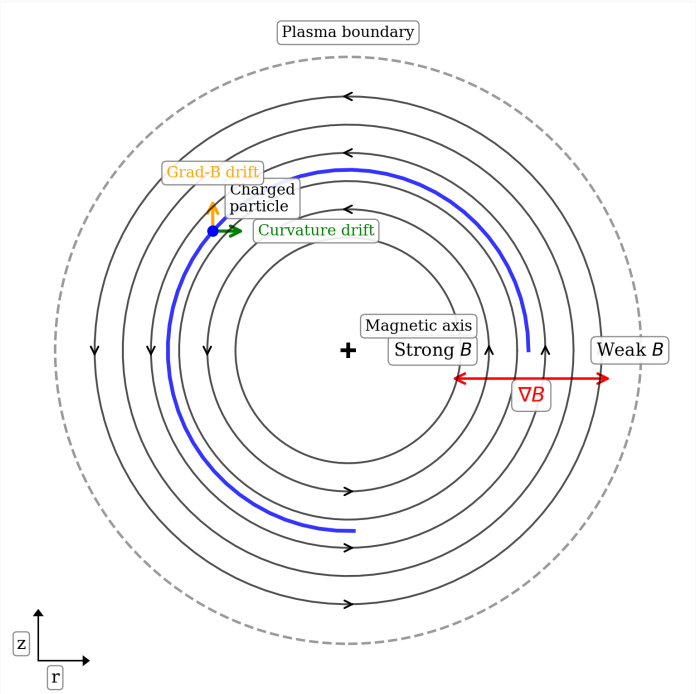

Fig. 3. Diagram for Question 3

Question 4

10.0 points

Examine the q-profile (safety factor) and magnetic flux surfaces in the tokamak shown in the diagram below. Discuss how the shape of the q-profile influences plasma stability and confinement.

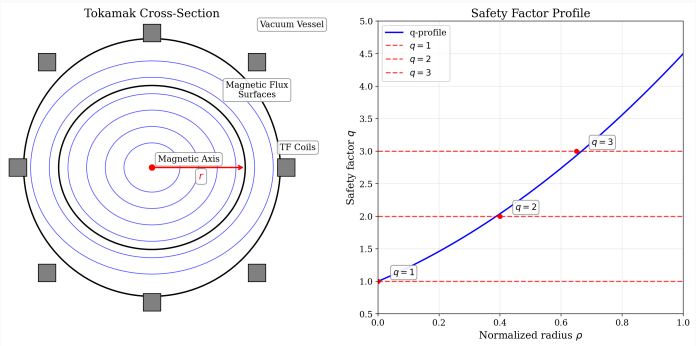

Fig. 4. Diagram for Question 4

Question 5

10.0 points

Examine the diagram below illustrating plasma instability growth rate as a function of wave number. Discuss the implications of this relationship for tokamak operation and stability control.

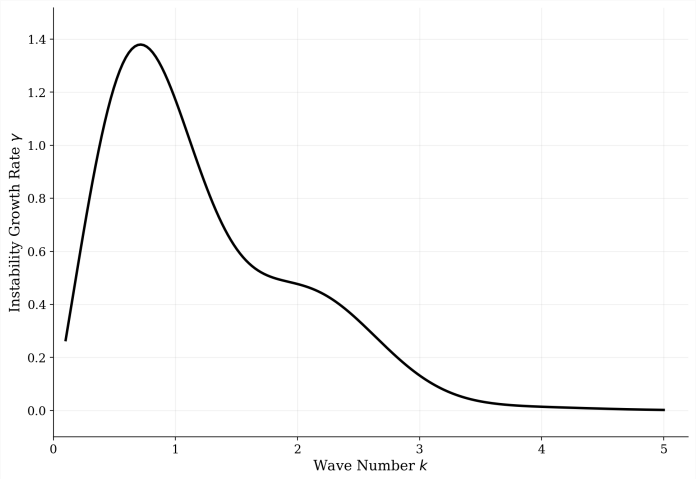

Fig. 5. Diagram for Question 5